

Machine Learning and its Applications

Today's goals

Module 1

Machine Learning Fundamentals I: Basic Concepts

Module 2

Machine Learning Fundamentals II: Model Training and Evaluation

Module 3

Data Preparation

Module 9 Introduction

Introduction to Module 9's Study

Video • 2 min

Why Data Preparation Is Not Optional in

Machine Learning

Reading • 5 min

Module 9-1 Basic Data Cleaning Review

Basic Data Cleaning Review

Video • 38 sec

Materials on Basic Data Cleaning

Reading • 10 min

Cleaning Data Is a Decision, Not a Checklist

Reading • 5 min

Module 9-2 Distributions, Outliers and Their Removal

Distributions, Outliers and Their Removal: Gaussian Distribution, Skewness and Outliers

Video • 6 min

Distributions, Outliers and Their Removal: The Z-Score and IQR Method for Outliers Removal

Video • 5 min

Materials on Distributions, Outliers and Their Removal

Reading • 10 min

Outliers, Assumptions, and When “Cleaning” Becomes Damage

Reading • 5 min

Module 9-3 Data Transform

Data Transform: Normalization, Standardization, Power Transform

Video • 7 min

Materials on Data Transform

Reading • 10 min

When Scaling Helps — and When It Quietly Breaks Your Model

Reading • 5 min

Module 9-4 Construct Training and Test Sets for Model Evaluation

Construct Training and Test Sets for Model Evaluation: Methods, Implementation, and Stratified Sampling

Video • 5 min

Construct Training and Test Sets for Model Evaluation: Cross-Validation

Video • 6 min

Materials on Training/Test Sets Generation

Reading • 10 min

Evaluation Starts with the Split: When Performance Numbers Lie

Reading • 5 min

Module 9-5 Data Preparation Overview

Data Preparation Overview

Video • 3 min

Materials on Data Preparation

Reading • 10 min

From Raw Data to Reliable Learning: What Actually Matters?

Reading • 5 min

Data Preparation and Evaluation Coach Dialogue

• 5 min

Module 9 Quiz

Module 9 Quiz

Practice Assignment • 30 min

Module 9 Assignments

Module 9 Assignment 1

Graded App Item • 30 min

Module 9 Assignment 2

Graded App Item • 30 min

Module 9 Assignment 3

Graded App Item • 30 min

XLS file for Module 9 Assignment

Reading • 1 min

Module 9 Recap

Reading • 5 min

Module 4

Support Vector Machines

Module 10 Introduction

Introduction to Module 10's Study

Video • 1 min

Why Support Vector Machines?

Reading • 5 min

Module 10-1 Support Vector Machine Fundamentals

Support Vector Machine Fundamentals: Concepts

# Evaluation Starts with the Split: When Performance Numbers Lie



## 1. Reflecting on the Importance of Independence

- Reflect on why the test dataset must remain completely independent from the training dataset when evaluating a machine learning model.
- Consider the types of mistakes that could unintentionally break this independence, even when a train/test split appears to have been performed correctly.

## 2. Reflecting on Single Splits vs. Cross-Validation

- Reflect on the limitations of evaluating a machine learning model using only a single train–test split.
- Consider why cross-validation may provide a more reliable and stable estimate of model performance across different subsets of data.

## 3. Reflecting on Stratified Sampling

- In classification problems, reflect on why purely random sampling could lead to misleading or unstable evaluation results, especially when class distributions are imbalanced.
- Consider how stratified sampling may help preserve class distributions and reduce bias or variance during model evaluation.

## 4. Reflecting on Choosing an Evaluation Strategy

- Reflect on situations where a simple train–test split may be sufficient for evaluating a model.
- Consider when K-fold cross-validation or leave-one-out cross-validation may be more appropriate.
- Reflect on the trade-offs involved between computational cost, training time, and the reliability of evaluation results.

## 5. Reflecting on Data Preparation and Splitting

- Reflect on how preprocessing steps such as outlier removal, scaling, normalization, or transformations may interact with train/test splitting.
- Consider why certain preprocessing operations should be performed only after splitting the data in order to avoid data leakage and misleading evaluation outcomes.

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